

National Exam of Higher National Diploma-New program – 2020 Session

Spécialty/option : SWE-CSN-DBM

Paper : Digital Electronics

Duration : 4 hours



Credit : 7

*Instructions: Answer all questions. You are authorized to use only non-programmable calculator.*

**SECTION A: NUMBER SYSTEM AND CODES. Each question carries 20 marks.**

1. In the decimal numbering system, what is the MSD?
  - A. The middle digit of a stream of numbers
  - B. The digit to the right of the decimal point
  - C. The last digit on the right
  - D. The digit with the most weight
2. The output of an AND gate with three inputs, A, B, and C, is HIGH when \_\_\_\_\_.
  - A.  $A = 1, B = 1, C = 0$
  - B.  $A = 0, B = 0, C = 0$
  - C.  $A = 1, B = 1, C = 1$
  - D.  $A = 1, B = 0, C = 1$
3. The BCD number for decimal 16 is \_\_\_\_\_.
  - A. 00010110
  - B. 00010000
  - C. 00010010
  - D. 11100000
4. What are the symbols used to represent digits in the binary number system?
  - A. 1,2
  - B. 0 through 8
  - C. 0,1,2
  - D. 0,1
5. Convert the fractional binary number 0001.0010 to decimal.
  - A. 1.40
  - B. 1.125
  - C. 1.20
  - D. 1.80
6. The number of digits used to store a BCD number is:
  - A. 1
  - B. 8
  - C. 4
  - D. 2

7. Convert  $11001001_2$  (binary) to decimal.
- A. 201
  - B. 2001
  - C. 210
  - D. 20
8.  $17010$  is equivalent to
- A.  $FD_{16}$
  - B.  $DF_{16}$
  - C.  $AA_{16}$
  - D.  $AF_{16}$
9. The operation  $111000 + 110110$  in binary is equal to
- A. 111111
  - B. 1101110
  - C. 1111110
  - D. 1011110
10. The operation  $111000 - 101111$  in binary is equal to
- A.  $11_{10}$
  - B.  $10_{10}$
  - C.  $9_{10}$
  - D.  $8_{10}$
11. The two's complement of binary number 0101 is
- A. 1111
  - B. 1101
  - C. 1110
  - D. 1011
12. Which of the following logical operations is represented by the + sign in Boolean algebra?
- A. inversion
  - B. AND
  - C. OR
  - D. complementation
13. The output of an AND gate is LOW \_\_\_\_\_.
- A. all the time
  - B. when any input is LOW
  - C. when any input is HIGH
  - D. when all inputs are HIGH
14. Excess-3 code is known as :
- A. Weighted code
  - B. Cyclic redundancy code
  - C. Self-complementing code
  - D. Algebraic code
15. A full adder logic circuit will have
- A. Two inputs and one output.
  - B. Three inputs and three outputs.
  - C. Two inputs and two outputs.
  - D. Three inputs and two outputs.

16. How many AND gates are required to realize the circuit of the function  $Y = AD + AB + BC$

- A. 4
- B. 5
- C. 3
- D. 2

17. The 1's complement can be easily obtained by using \_\_\_\_\_

- A) Comparator
- B) Inverter
- C) Adder
- D) Subtractor

18. Any signed negative binary number is recognized by its \_\_\_\_\_

- A. MSB
- B. LSB
- C. Byte
- D. Nibble

19. If the decimal number is a fraction then its binary equivalent is obtained by \_\_\_\_\_ the number

continuously by 2.

- A. Dividing
- B. Multiplying
- C. Adding
- D. Subtracting

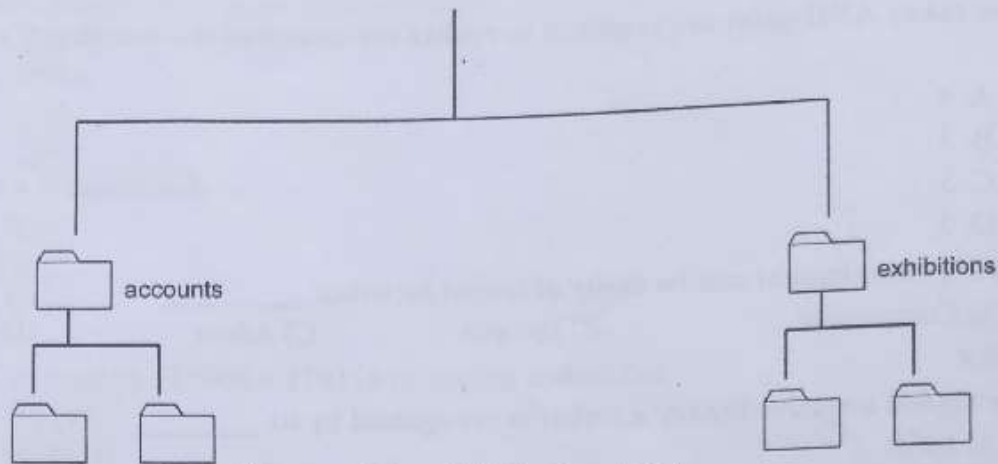
20. The format used to present the logic output for the various combinations of logic inputs to a gate is called a(n):

- A. Boolean algebra
- B. Boolean variable
- C. Truth table
- D. Logic function

## **SECTION B: COMPUTER FUNDAMENTAL (80%)**

### **I- HARDWARE (20MARKS)**

1. Compare a hardware and a software (2marks)
2. Give 3 examples of computer input devices and 3 examples of output devices. For each state the use. (3marks)
3. Using a diagram explain the computer memory hierarchy (2marks)
4. What is a bus? (1marks)
5. List the different types of ROM. Which type is used in Flash technology? (2marks)
6. Explain serial and parallel transmission of data (2marks)
7. What is the motherboard? Give 5 devices attached to the motherboard (3marks)
8. Trendy Artists gallery has a directory structure with the C drive as its root directory.



Kim creates templates that will be used for exhibition brochures.

- Redraw the above diagram and indicate where the brochure templates for the brochures should be saved. **(1marks)**
- What is the absolute directory path for the brochure templates? **(1marks)**
- How can she protect the brochure templates from being modified? **(1marks)**
- Kim wants to have all file names with directory location shown automatically on the bottom of each page. A macro has been suggested.
- Should this be a local or global macro? Justify your answer. **(2marks)**

## II- NETWORK AND MOBILE DEVICES (40MARKS)

- What are the types of mobile applications? **(3mks)**
- What does WAP mean? **(2mks)**
- What is GPRS and how does it work? **(3mks)**
- What is the role of the following in the internet communication? **(10mks)**
  - Modem
  - Network Server
  - DNS Server
  - E-mail
  - Videoconference
- Through which means can a wireless communication take place? **(3mks)**
- With the aid of diagrams state and explain the different types of computer network topology and which topology do you consider the best **(6marks)**
- What is the function of the following networking devices **(10marks)**
  - Distributor
  - Router
  - Switch
  - NIC
  - Network cable
- What is a mobile device? **(1mark)**
- List the different types of mobile devices you know **(2marks)**

### III- MICROSOFT WORD, EXCEL AND POWERPOINT (20MARKS)

1. What is the suitable Office software to perform the following actions? (4mks)

- a) Write a letter
- b) Write a financial report
- c) Design a calendar
- d) Prepare a presentation for a seminar

2. What is the use of the following Microsoft Excel functions? (3mks)

- a) MAX
- b) AVERAGE
- c) SUM

3. From the figure below:

|   | A               | B    | C      | D     | E |
|---|-----------------|------|--------|-------|---|
| 1 |                 | Bill |        |       |   |
| 2 | Name            | Unit | Number | Total |   |
| 3 | Iron (8)        | 2875 | 450    |       |   |
| 4 | Iron (10)       | 4590 | 25     |       |   |
| 5 | Cement          | 5900 | 350    |       |   |
| 6 | Corrugated Iron | 6300 | 600    |       |   |
| 7 |                 |      | Total  |       |   |

- a) From which application software is this figure? (1 mark)
- b) Give the formula to be used to fill the cells D3, D4, D5, D6 and D7 (5 marks)
- c) How can you from D3, automatically fill D4-D6? (1 marks)

- 4. What is the importance of applying breaks in a document? Give the procedures (2marks)
- 5. What is the file extension for a word, excel and PowerPoint document. (2marks)
- 6. What is the procedure for creating animations and transition in PowerPoint (2marks)